CSSE 220 – Object-Oriented Software Development
Rose-Hulman Institute of Technology

Worksheet 14

Name (Print):_____ Section:_____

1. **helper method recursion**: The _____ method starts the recursion, and the _____ method handles the recursive logic.

2. Complete a Java program that prints all numbers from 1 up to n, using recursion and a helper method.

```
public static void countUp(int n){
// TODO: call the helper method to start recursion
  ---------------
  }
private static void countUpHelper(int current, int end){
// TODO: base case
  ---------------
  System.out.println(current);
// TODO: recursive case
  -----------------;
}
public static void main(String[] args) {
     countUp(5); // prints 1,2,3,4,5
  }
```

3. **Polymorphism Basics**. Consider the following code:

```
Enemy e1 = new Zombie();
e1.attack();
```

   (a) The **reference type** of `e1` is: _____

   (b) The **object type** created at run-time is: _____

   (c) Which type determines whether `attack()` is a **legal method call**? _____

   (d) Which type determines **which version of `attack()` runs**? _____

4. **Compile-Time vs Run-Time**

   (a) _____: checks whether a method call is allowed

   (b) _____: decides which overridden method executes

   (c) _____: based on the reference type

   (d) _____: based on the actual object

5.

```
interface Top {
    public void alpha();
    public void beta();
    public void gamma();
    public void delta();
    // Note no epsilon here
}
```

```
class One implements Top {

    public void alpha() {
        System.out.println("A");
    }

    public void beta() {
        System.out.println("B");
    }

    public void gamma() {
        System.out.println("C");
    }

    public void delta() {
        System.out.println("D");
        this.beta();
    }
}
```

```
class Two extends One
        implements Top {

    public void beta() {
        System.out.println("E");
    }

    public void gamma() {
        super.gamma();
        System.out.println("F");
    }

    public void epsilon() {
        System.out.println("G");
    }
}
```

Two m = new Two();
Top q = new One();
Top r = new Two();
One s = new Two();

Code	Output Choices (circle one in each problem)															
														no output	*runtime error*	*compile error*
`m.alpha();`	A	B	C	D	E	F	G	BE	CF	DB	DE	EB	FC			
														no output	*runtime error*	*compile error*
`m.gamma();`	A	B	C	D	E	F	G	BE	CF	DB	DE	EB	FC			
														no output	*runtime error*	*compile error*
`m.omega();`	A	B	C	D	E	F	G	BE	CF	DB	DE	EB	FC			
														no output	*runtime error*	*compile error*
`q.alpha();`	A	B	C	D	E	F	G	BE	CF	DB	DE	EB	FC			
														no output	*runtime error*	*compile error*
`r.beta();`	A	B	C	D	E	F	G	BE	CF	DB	DE	EB	FC			
														no output	*runtime error*	*compile error*
`r.epsilon();`	A	B	C	D	E	F	G	BE	CF	DB	DE	EB	FC			
														no output	*runtime error*	*compile error*
`s.beta();`	A	B	C	D	E	F	G	BE	CF	DB	DE	EB	FC			